

THEORY OF COMPUTATION (CS302)

5 Marks Question Bank

UNIT 1 – DFA & NFA

1. Define Deterministic Finite Automata (DFA).
2. Define Non-Deterministic Finite Automata (NFA).
3. Differentiate DFA and NFA.
4. Explain the components of a finite automaton.
5. Construct a DFA for strings ending with '01'.
6. What are epsilon transitions?

UNIT 2 – Regular Languages

1. Define Regular Expression.
2. Explain the operators used in Regular Expressions.
3. What is Pumping Lemma?
4. State the Pumping Lemma for Regular Languages.
5. Write short notes on Regular Languages.
6. Explain applications of Regular Expressions.

UNIT 3 – Context-Free Grammars

1. Define Context-Free Grammar (CFG).
2. Explain the components of a CFG.
3. Differentiate terminals and non-terminals.
4. What is a Parse Tree?
5. Define Ambiguous Grammar.
6. Explain Chomsky Normal Form (CNF).

UNIT 4 – Pushdown Automata

1. Define Pushdown Automata (PDA).
2. Explain the components of PDA.
3. What are stack operations in PDA?
4. Differentiate acceptance by final state and empty stack.
5. Explain the relationship between PDA and CFG.
6. Write short notes on Instantaneous Description (ID).

UNIT 5 – Turing Machines & Decidability

1. Define Turing Machine.
2. Explain the components of a Turing Machine.
3. What is Decidability?
4. Define Recursive and Recursively Enumerable Languages.
5. Explain the Halting Problem.
6. State the Church-Turing Thesis.

Exam Preparation Tips

Focus on DFA/NFA conversions, Regular Expressions, Pumping Lemma, CFGs, CNF, PDA constructions, Turing Machines, and Decidability concepts as they are frequently asked in university examinations.